


FYS-STK3155/4155 September 14

Momentum

$$m \frac{d^2 x}{dt^2} + \mu \frac{dx}{dt} = - \frac{dV(x)}{dx}$$

$$\frac{d^2 x}{dt^2} \approx \frac{x_{t+\Delta t} + x_{t-\Delta t} - 2x_t}{(\Delta t)^2}$$

$$\frac{dx}{dt} \approx \frac{x_{t+\Delta t} - x_t}{\Delta t}$$

$$\Delta x_{t+\Delta t} = x_{t+\Delta t} - x_t$$

$$\Delta x_t = x_t - x_{t-\Delta t}$$

$$\frac{m}{\Delta t^2} \Delta x_{t+\Delta t} - \frac{m}{\Delta t^2} \Delta x_t$$

$$+ \frac{\mu}{\Delta t} \Delta x_{t+\Delta t} = - \frac{dV}{dx}$$

$$\Delta x_{t+\Delta t} = \Delta x_t \left(\frac{m}{m + \mu \Delta t} \right)^{\beta}$$

$$- \frac{dV}{dx} \left(\frac{\Delta t^2}{m + \mu \Delta t} \right)^{\gamma}$$

$$\Delta X_{t+\Delta t} = \Delta X_t \cdot \beta - \delta q$$

$$\lim_{n \rightarrow \infty} \beta = \underline{1} \quad \frac{dv}{dx}$$

$$\lim_{n \rightarrow \infty} \beta = 0 \quad \beta \in [0, 1]$$

$$X_t \rightarrow \ominus^{(n)}$$

$$X_{t+\Delta t} \rightarrow \ominus^{(n+1)}$$

$$X_{t-\Delta t} \rightarrow \ominus^{(n-1)}$$

$$E^{(n+1)} = E^{(n)} - \alpha g(E^{(n)}) \\ + \beta [E^{(n)} - E^{(n-1)}]$$

$$\beta \in [0, 1]$$

introduces some
memory in terms of the
value $E^{(n-1)}$

algorithm

- initialize $\epsilon^{(0)}$
- fix γ
- fix β
- initialize $v^{(0)}$

while stopping criterion not met

$$v^{(n)} = \beta (\epsilon^{(n)} - \epsilon^{(n-1)}) - g(\epsilon^{(n)}) \cdot \gamma$$

$$\epsilon^{(n+1)} = \epsilon^{(n)} + v^{(n)}$$

end while

Automatic differentiation
— Repeated application of
the chain on elementary
functions. Could be
a program which we
can break into elemen-
tary functions.

Back propagation algo
for deep learning is
nothing but AD
in Reverse mode,

Example 1 2 Flops

$$f(x) = \exp(x^2) = b \\ = \exp(a) ; a = x^2$$

$$f'(x) = 2x \cdot \exp(x^2) \quad 4 \text{ Flops} \\ = 2x \cdot b \quad 2 \text{ Flops}$$

$$\frac{df}{dx}$$

=

$$\begin{bmatrix} \frac{df}{db} & \frac{db}{da} & \frac{da}{dx} \end{bmatrix}$$

$\underbrace{\hspace{1.5cm}}$

= 1

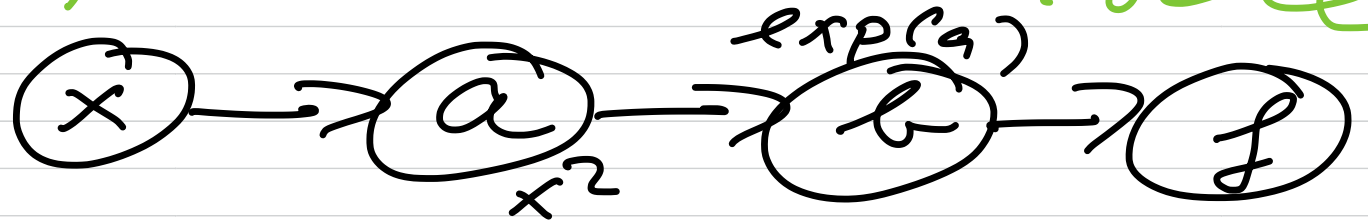
$\underbrace{\hspace{1cm}}$

$\exp(a)$

$\underbrace{\hspace{1cm}}$
b

$-2x$

Forward mode



Reverse mode

Begin with output
 $b = f = \exp(x^2)$

Example 2

$$f(x) = \sqrt{x^2 + \exp(x^2)}$$

5 FLOPS

$$f'(x) = \frac{x(1 + \exp(x^2))}{\sqrt{x^2 + \exp(x^2)}}$$

10 FLOPS

$$a = x^2 \quad \wedge \quad b = \exp(a)$$

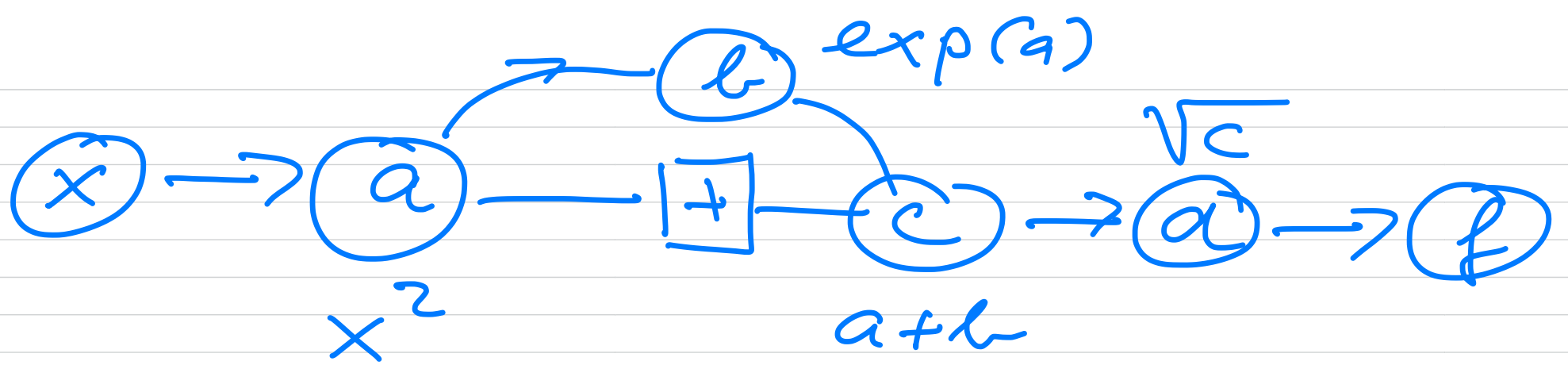
$$c = a + b \quad \wedge \quad d = \sqrt{c}$$

$$= \sqrt{a+b} = \sqrt{x^2 + \exp(x^2)}$$

$$= f(x)$$

$$f'(x) = \frac{x(1+b)}{\sqrt{c}}$$

3 FLOPs -



$\frac{\partial f}{\partial x}$ by working backwards
(reverse mode)

$$\frac{\partial f}{\partial d} = 1 \quad f(x) = d = \sqrt{c}$$

$$\frac{\partial f}{\partial c} = \frac{\partial f}{\partial d} \frac{\partial d}{\partial c} = \frac{1}{2\sqrt{c}}$$

$$\frac{\partial f}{\partial b} = \frac{\partial f}{\partial c} \frac{\partial c}{\partial b} = \frac{1}{2\sqrt{c}}$$

$$c = a + b \quad \frac{\partial c}{\partial b} = 1$$

$$\frac{\partial f}{\partial a} = \frac{\partial f}{\partial b} \frac{\partial b}{\partial a} + \frac{\partial f}{\partial c} \frac{\partial c}{\partial a}$$

$\frac{\partial b}{\partial a} = 1 \quad \frac{\partial c}{\partial a} = 1$

$$= \frac{1 + b}{2\sqrt{c}}$$

$$\left| \begin{array}{l} \frac{\partial f}{\partial x} = \frac{\partial f}{\partial a} \frac{\partial a}{\partial x} \\ a = x^2 \end{array} \right.$$

$$= \frac{(1+b)x}{\sqrt{c}} = f'(x)$$

$$f(x) = \sqrt{c}$$

$$= \sqrt{a+b}$$

$$= \sqrt{x^2 + \exp(a)}$$

Let $x_1, \dots, x_d$ be input
variables ($x_1 = x$)
to the function f -

($d=1$ $x_1 = x$)

$x_{d+1}, \dots, x_D$ intermediate
variables - Here $x_D = x_5$

x_D = output variable

$$x_2 = a \Rightarrow \boxed{x^2} = x_1^2 \wedge x_3 = b = \exp(x_2)$$

$$x_4 = a + b = c \wedge x_5 = d = \sqrt{c} = f$$

$$x_i' = g_i(x_{pa}(x_i))$$

$$x_3 = b = \underbrace{\exp}_{g_3}(a)$$

g_i represents elementary functions and $x_{pa}(x_i)$ are parent nodes of the variable x_i' in comp. graph.

$$\frac{\partial f}{\partial x_i} = \sum_{x_j: x_i \in P_a(x_j)} \frac{\partial f}{\partial x_j'} \frac{\partial x_j'}{\partial x_i'}$$

$$= \sum_{x_j': x_i \in P_a(x_j')} \frac{\partial f}{\partial x_j} \frac{\partial g_j'}{\partial x_i'}$$